

# STORYTELLING WITH DATA VISUALIZATION

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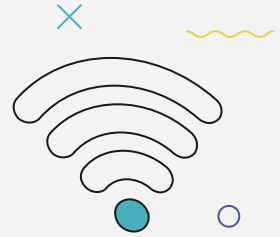
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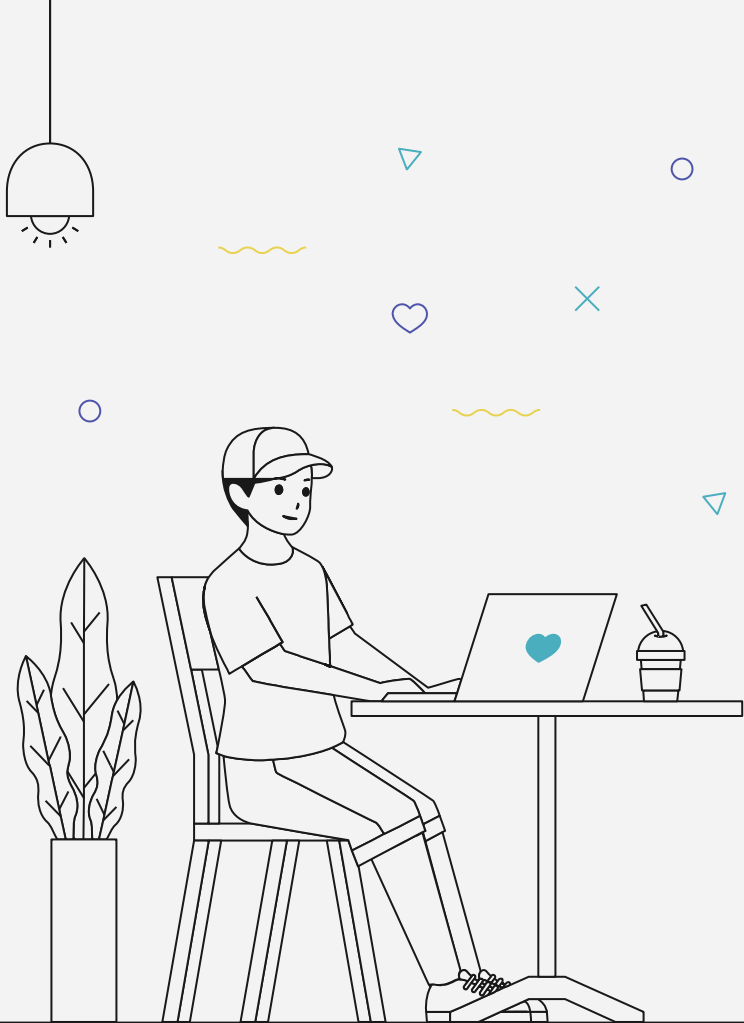
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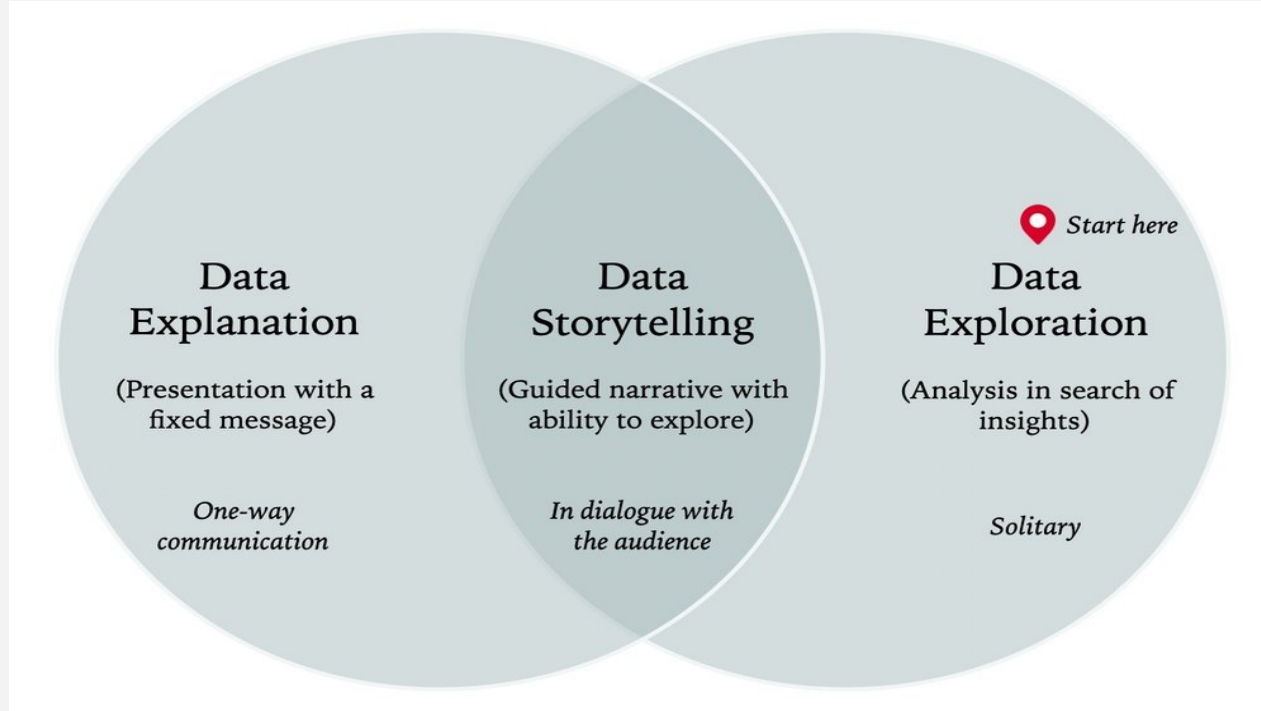
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# 01 The importance of context



# Exploratory vs. explanatory analysis





# Explanatory analysis



## WHO

whom are you communicating?



## WHAT

What do you want your audience to know or do?



## HOW

How can you use data to help make your point?





# The 3-minute story & Big Idea

## 3-minute story

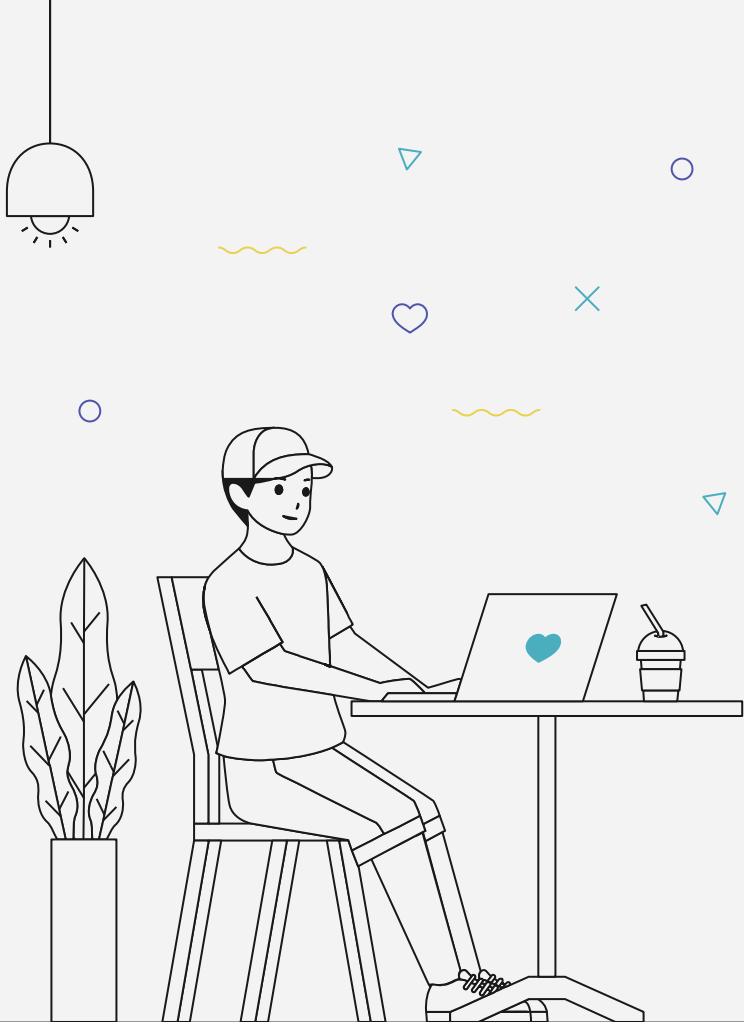
If you had only three minutes to tell your audience what they need to know, what would you say?

## Big Idea

Big Idea has **THREE** components:

1. It should clearly show your own opinion or perspective.
  2. It should explain why it matters or what's at risk.
  3. It must be a complete sentence
- 
- 
- 
- 

# 02 Constructing the story



# The beginning

The first thing to do is introduce the plot, building the context for your audience

Why should I  
pay  
attention?

What is in it  
for me?



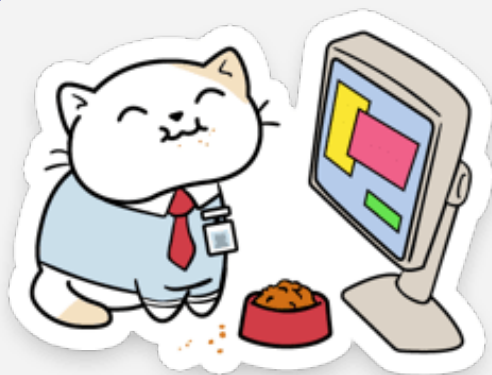
1. **The setting** : When and where does the story take place?
2. **The main character** : Who is driving the action? (This should be framed in terms of your audience!)
3. **The imbalance** : Why is it necessary, what has changed?
4. **The balance** : What do you want to see happen?
5. **The solution** : How will you bring about the changes?



# The middle

Focus on showing the problem clearly, explain your solution, and convince your audience why it's the right action to take.

- Further develop the situation or problem by covering relevant background.
- Incorporate external context or comparison points.
- Give examples that illustrate the issue.
- Include data that demonstrates the problem.
- Articulate what will happen if no action is taken or no change is made.
- Discuss potential options for addressing the problem.
- Illustrate the benefits of your recommended solution.
- Make it clear to your audience why they are in a

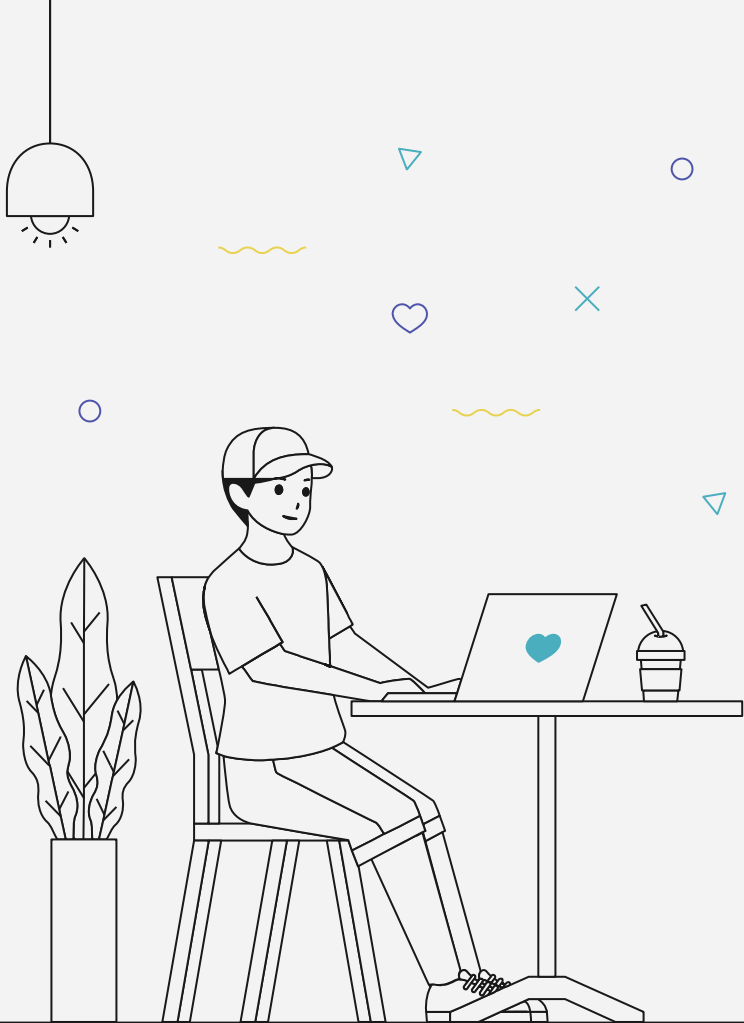


# The end

End with a call to action: make it totally clear to your audience what you want them to do with the new understanding or knowledge that you've imparted to them



# 03 Choosing an effective visual



# How do We visualize data?



We first need to look at two things:

- Types of data
- Preattentive attributes





# Types of data

(nominal data)

## Categorical Data

The values of a **nominal attribute** are symbols or *names of things*.

## Binary Attributes

A **binary attribute** is a nominal attribute with only two categories or states:

- **0** means that the attribute is absent
- **1** means that it is present.

## Ordinal Attributes

An **ordinal attribute** is an attribute with possible values that have a meaningful order or *ranking* among them, but the magnitude between successive values is not known

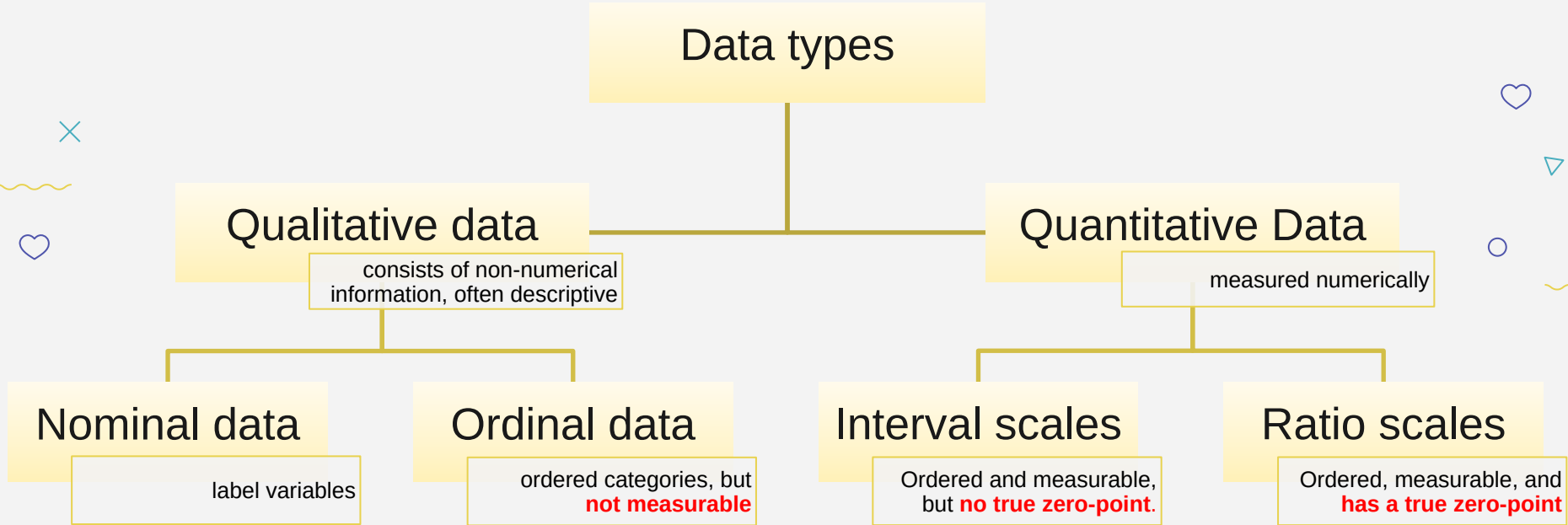
(numerical data)

## Quantitative Data

A **numeric attribute** is *quantitative*; that is, it is a measurable quantity, represented in integer or real values



# Types of data



# Preattentive attributes

We need to look at the different types of data, in order to choose the best visual encoding for each type.



|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 7 | 1 | 1 | 6 | 9 | 1 |
| 9 | 1 | 7 | 5 | 5 | 5 | 6 | 2 | 5 | 9 |
| 4 | 5 | 2 | 9 | 6 | 9 | 7 | 6 | 4 | 6 |
| 8 | 1 | 5 | 7 | 8 | 5 | 6 | 6 | 6 | 7 |
| 7 | 2 | 3 | 6 | 8 | 9 | 1 | 7 | 9 | 1 |
| 3 | 8 | 6 | 8 | 4 | 5 | 6 | 9 | 4 | 5 |
| 4 | 9 | 9 | 2 | 3 | 7 | 1 | 9 | 1 | 2 |
| 3 | 7 | 8 | 1 | 6 | 1 | 5 | 6 | 1 | 6 |
| 5 | 6 | 6 | 8 | 6 | 6 | 9 | 1 | 2 | 6 |
| 3 | 2 | 4 | 2 | 6 | 9 | 4 | 2 | 7 | 1 |

FIGURE 1.3 How many 9s are there?

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 7 | 1 | 1 | 6 | 9 | 1 |
| 9 | 1 | 7 | 5 | 5 | 5 | 6 | 2 | 5 | 9 |
| 4 | 5 | 2 | 9 | 6 | 9 | 7 | 6 | 4 | 6 |
| 8 | 1 | 5 | 7 | 8 | 5 | 6 | 6 | 6 | 7 |
| 7 | 2 | 3 | 6 | 8 | 9 | 1 | 7 | 9 | 1 |
| 3 | 8 | 6 | 8 | 4 | 5 | 6 | 9 | 4 | 5 |
| 4 | 9 | 9 | 2 | 3 | 7 | 1 | 9 | 1 | 2 |
| 3 | 7 | 8 | 1 | 6 | 1 | 5 | 6 | 1 | 6 |
| 5 | 6 | 6 | 8 | 6 | 6 | 9 | 1 | 2 | 6 |
| 3 | 2 | 4 | 2 | 6 | 9 | 4 | 2 | 7 | 1 |

FIGURE 1.4 Now it's easy to count the 9s.

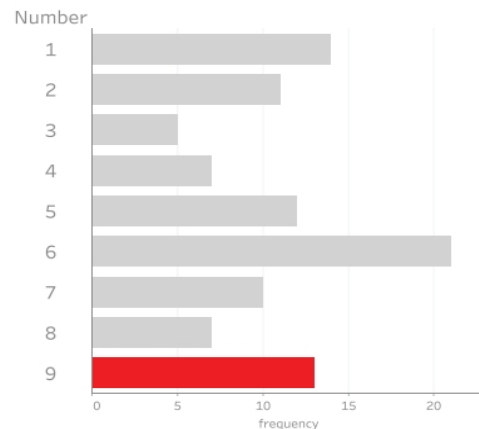


FIGURE 1.8 There are 13 9s.

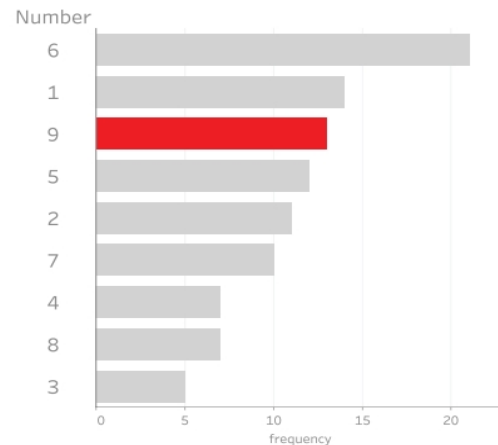
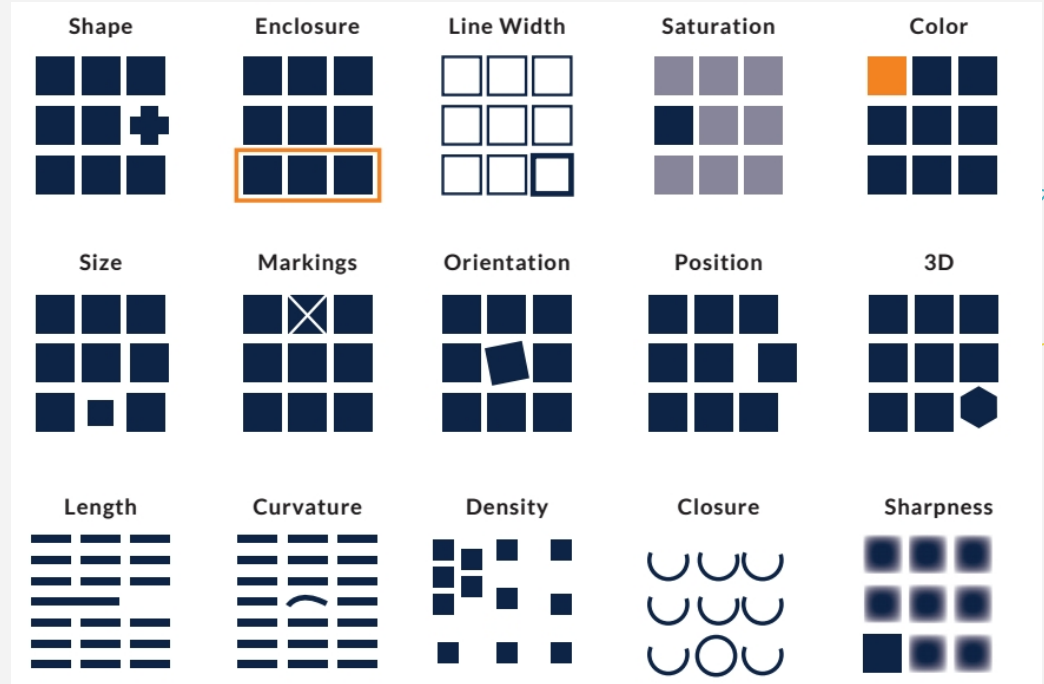


FIGURE 1.9 Sorted bar chart using color and length to show how many 9s are in our table.



# Preattentive attributes

We need to look at the different types of data, in order to choose the best visual encoding for each type.





# Preattentive attributes in text

❑ **Bold**, *italics*, and underlining: Use for titles, labels, captions, and short word sequences to differentiate elements

❑ **CASE and typeface**: Uppercase text in short word sequences is easily scanned, which can work well when applied to titles, labels and keywords.

❑ **Color** is an effective highlighting technique when used sparingly and generally in concert with other highlighting techniques

❑ **Inversing elements** is effective at attracting attention, but can add considerable noise to a design so should be used sparingly.

❑ **Size** is another way to attract attention and

## No preattentive attributes

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Color

What are we doing well? Great Products. **These products are clearly the best in their class.** Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Size

What are we doing well? Great Products. These products are the best in their class. Replacement parts are shipped when needed. You sent gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Outline (enclosure)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Bold

**What are we doing well?** Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Italics

What are we doing well? Great Products. These products are clearly the best in their class. *Replacement parts are shipped when needed.* You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Separate spatially

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Underline (added marks)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

FIGURE 4.5 Preattentive attributes in text



# Preattentive attributes in graph

## Top 10 design concerns

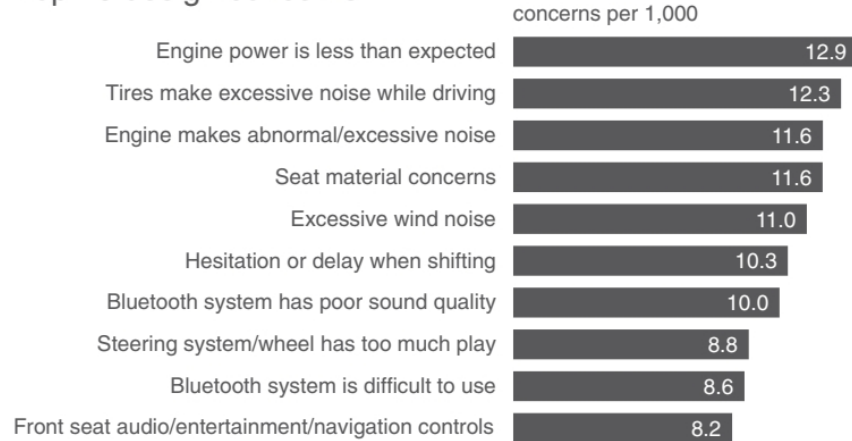


FIGURE 4.7 Original graph, no preattentive attributes

**7 of the top 10 design concerns have 10 or more concerns per 1,000.**

Discussion: is this an acceptable default rate?

## Top 10 design concerns

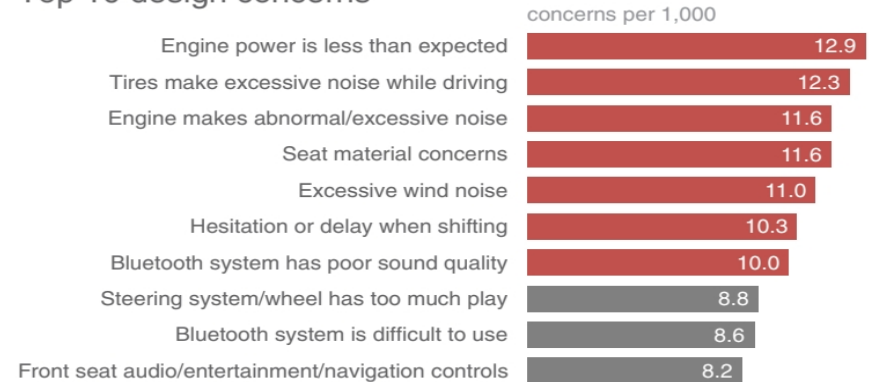


FIGURE 4.8 Leverage color to draw attention

# Encoding data in Charts

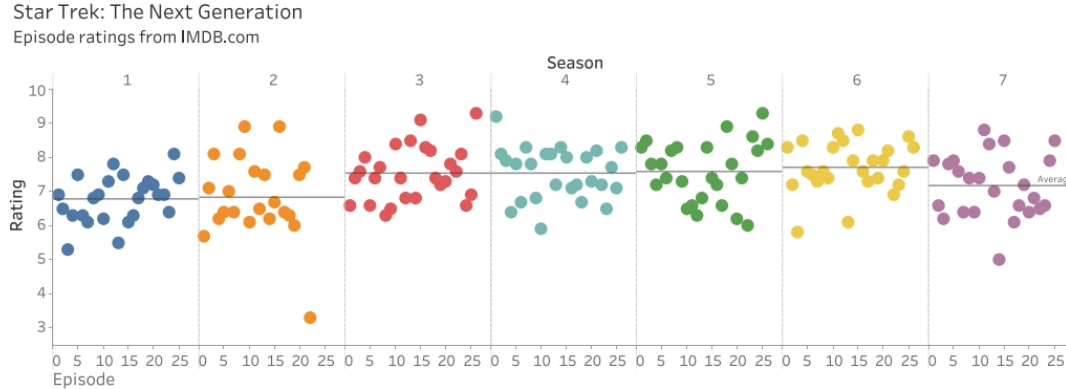


FIGURE 1.12 Every episode of *Star Trek: The Next Generation* rated.

Source: IMDB.com

TABLE 1.3 Data used in Figure 1.12.

| Data                  | Data Type    | Encoding | Note                                                                                                                                                                                                                                         |
|-----------------------|--------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Episode               | Categorical  | Position | Each episode is represented by a dot. Each dot has its own position on the canvas.                                                                                                                                                           |
| Episode Number        | Ordinal      | Position | The x-axis shows the number of each episode in each season.                                                                                                                                                                                  |
| Season                | Ordinal      | Color    | Each season is represented by a different color (hue).                                                                                                                                                                                       |
|                       |              | Position | Each season also has its own section on the chart.                                                                                                                                                                                           |
| IMDB rating           | Ordinal      | Position | The better the episode, the higher it is on the y-axis.                                                                                                                                                                                      |
| Average season rating | Quantitative | Position | The horizontal bar in each pane shows the average rating of the episodes in each season. There is some controversy over whether you should average ordinal ratings. We believe that the practice is so common with ratings it is acceptable. |

## Africa: The Malaria Center of the World

Africa continues to be the continent most seriously affected by malaria. While specific countries such as Kenya and São Tomé & Príncipe have all but eradicated deaths from the disease, death rates remain high in a large number of African countries.

Source: World Health Organization

### Malaria deaths in Africa, 2000-2014

#### The 17 most affected countries

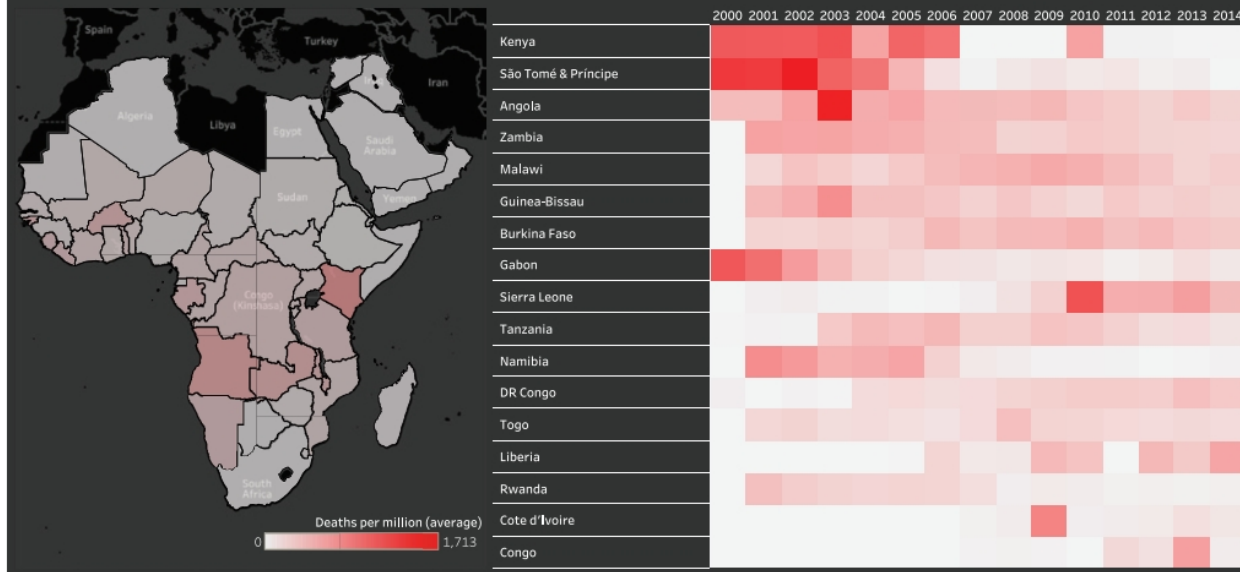


FIGURE 1.14 Deaths from malaria, 2000–2014.

Source: World Health Organization. Chart part of the Makeover Monday project

TABLE 1.5 Data used in the bar chart in Figure 1.14.

| Data               | Data Type    | Encoding | Note                                                                                              |
|--------------------|--------------|----------|---------------------------------------------------------------------------------------------------|
| Country            | Categorical  | Position | The map shows the position of each country. In the highlight table, each country has its own row. |
| Deaths per million | Quantitative | Color    | The map and table use the same color legend to show deaths per million people.                    |
| Year               | Ordinal      | Position | Each year is a discrete column in the table.                                                      |

Pick a Country

Nigeria

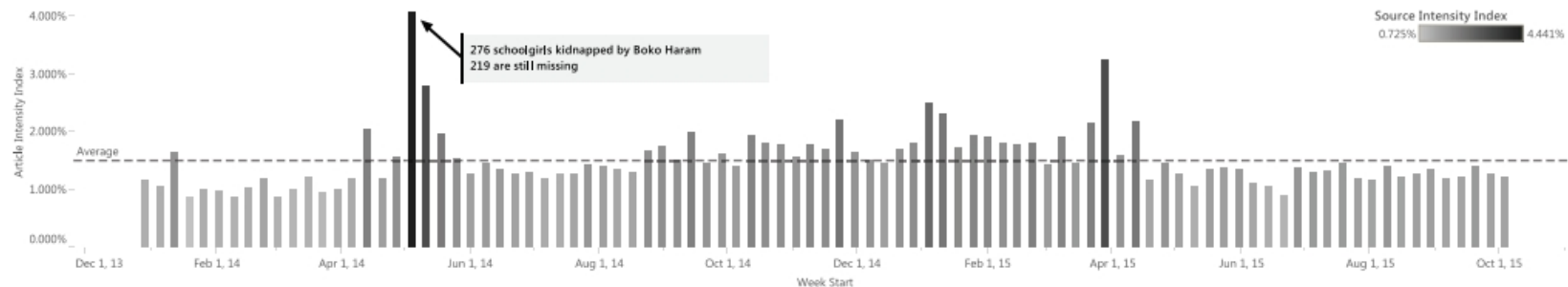
National News Special Edition 2014-2015

The Nigeria Weekly

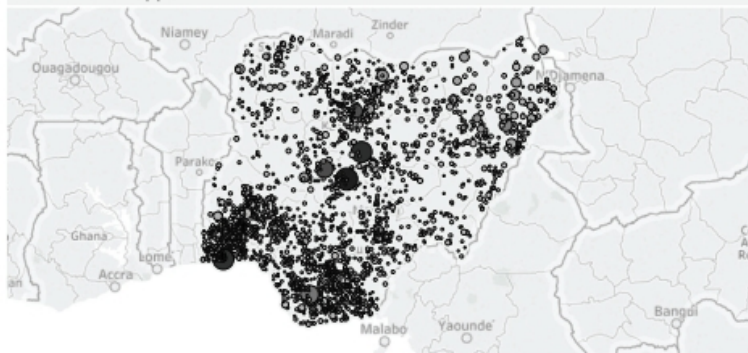
Monday, August 29, 2016 | Price: \$2000

Editor-in-Chief  
Shine Pulikathara  
@vizshine  
#ivpul

When did the key events in Nigeria happen? (click on bars to explore)



Where did it happen?



Who were talking about it?

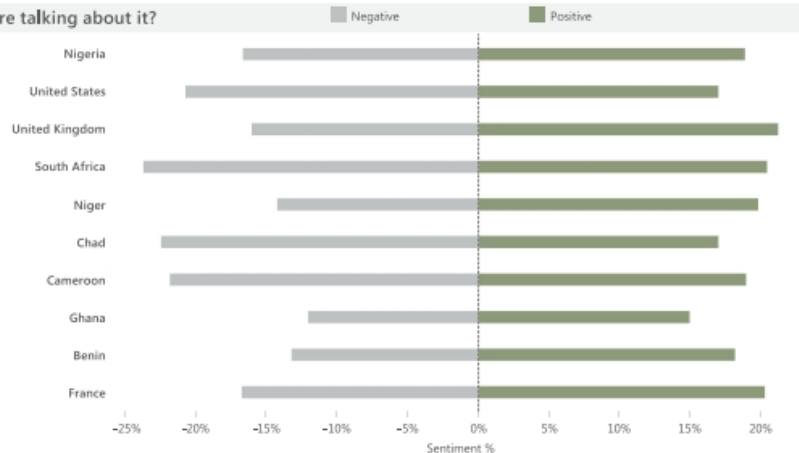


FIGURE 1.15 Winning visualization by Shine Pulikathara during the 2015 Tableau Iron Viz competition.

Source: Used with permission from Shine Pulikathara.

# Use of Color in Data Visualization

## SEQUENTIAL

color is ordered from low to high



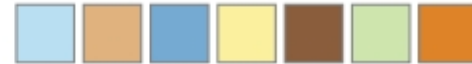
## DIVERGING

two sequential colors with a neutral midpoint



## CATEGORICAL

contrasting colors for individual comparison



## HIGHLIGHT

color used to highlight something



## ALERT

color used to alert or warn reader

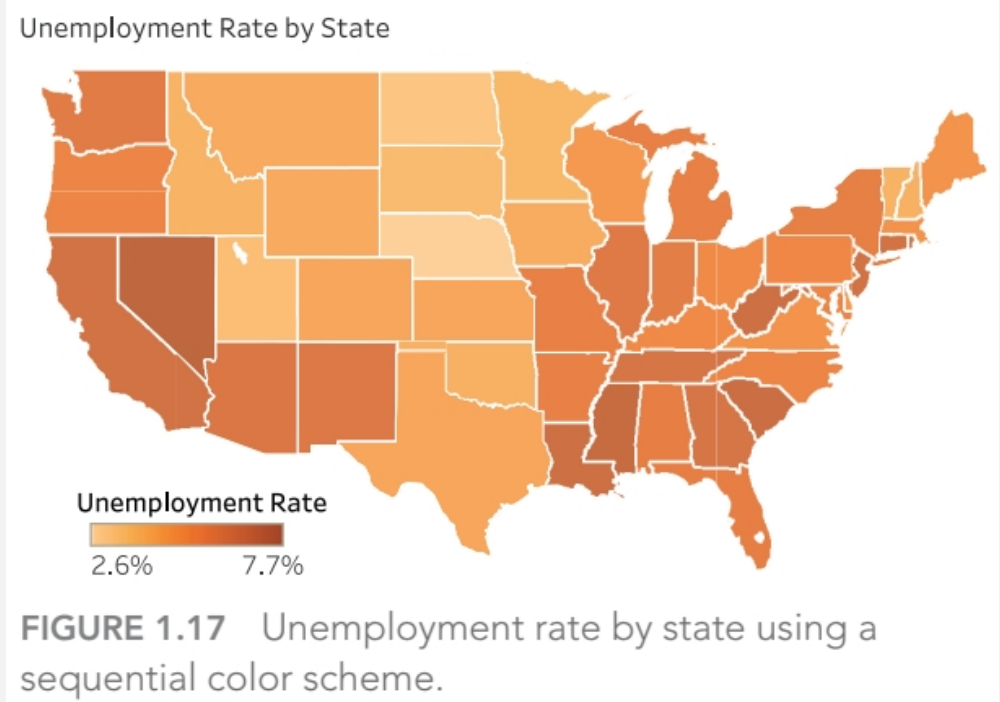


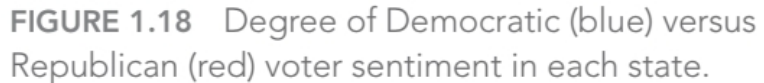
FIGURE 1.16 Use of color in data visualization.

# Use of Color in Data Visualization

**Sequential color** is the use of a single color from light to dark.

An example is encoding the total amount of sales by state in blue, where the darker blue shows higher sales and a lighter blue shows lower sales.





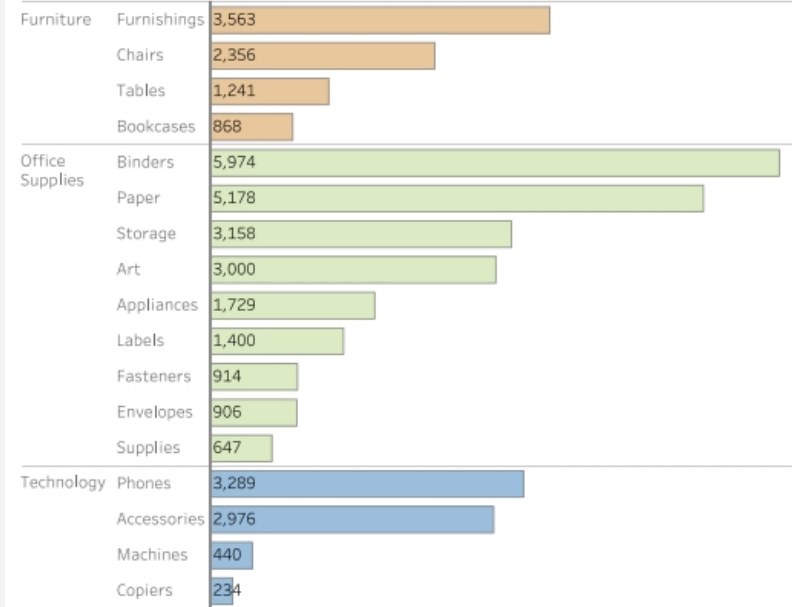


# Use of Color in Data Visualization

**Categorical color** uses different color hues to distinguish between different categories.

For example, we can establish categories involving apparel (e.g., shoes, socks, shirts, hats, and coats) or vehicle types (e.g., cars, minivans, sport utility vehicles, and motorcycles)

Quantity by Category



**FIGURE 1.20** Quantity of office supplies in three categories using a categorical color scheme.

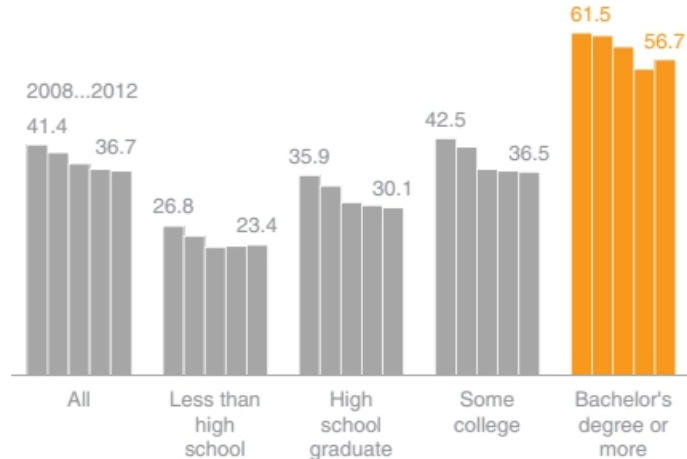
# Use of Color in Data Visualization

**Highlight color** is used when there is something that needs to stand out to the reader, but not alert or alarm them.

Highlights can be used in a number of ways, as in highlighting a certain data point, text in a table, a certain line on a line chart, or a specific bar in a bar chart

## New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

Source: U.S. Census

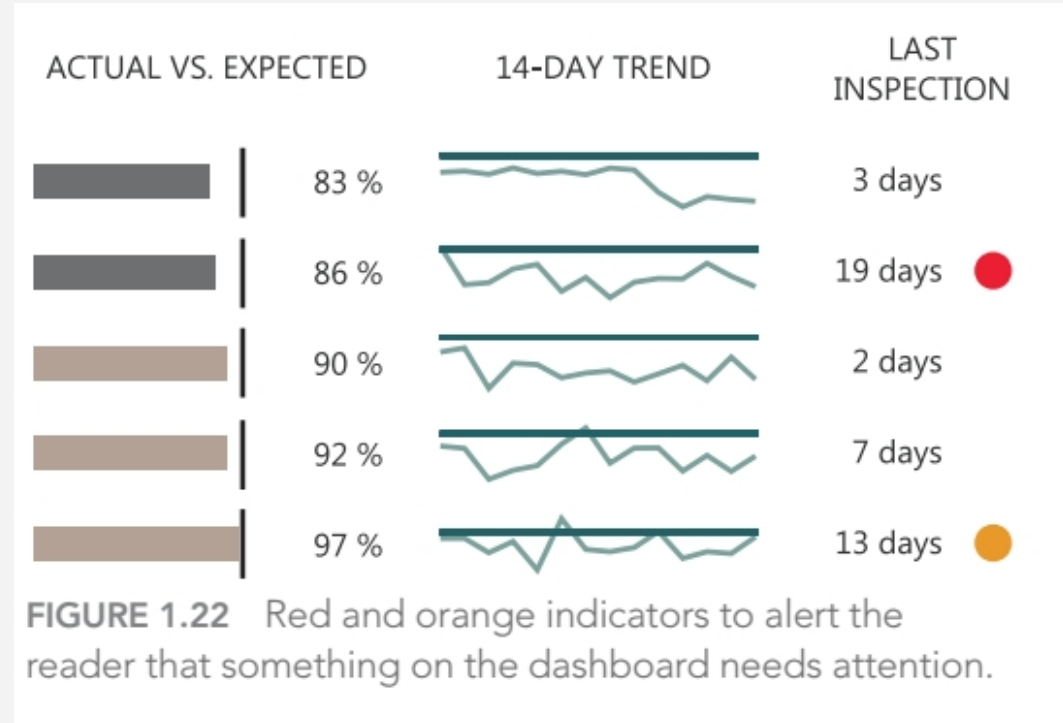
Adapted from PEW RESEARCH CENTER

FIGURE 5.3 Highlight the important stuff

# Use of Color in Data Visualization

**Alerting color** is used when there is a need to draw attention to something for the reader.

In this case, it's often best to use bright, alarming colors, which will quickly draw the reader's attention



# Use color sparingly and consistently

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

| Country | A | B | C  | D  | E  |
|---------|---|---|----|----|----|
| AUS     | 1 | 2 | 3  | 6  | 7  |
| BRA     | 1 | 3 | 4  | 5  | 6  |
| CAN     | 2 | 3 | 6  | 12 | 8  |
| CHI     | 1 | 2 | 8  | 4  | 7  |
| FRA     | 3 | 2 | 4  | 8  | 10 |
| GER     | 3 | 1 | 6  | 5  | 4  |
| IND     | 4 | 1 | 8  | 10 | 5  |
| ITA     | 2 | 4 | 10 | 9  | 8  |
| MEX     | 1 | 5 | 4  | 6  | 3  |
| RUS     | 4 | 3 | 7  | 9  | 12 |
| SPA     | 2 | 3 | 4  | 5  | 11 |
| TUR     | 7 | 2 | 3  | 4  | 8  |
| UK      | 1 | 2 | 3  | 6  | 7  |
| US      | 1 | 2 | 4  | 3  | 5  |

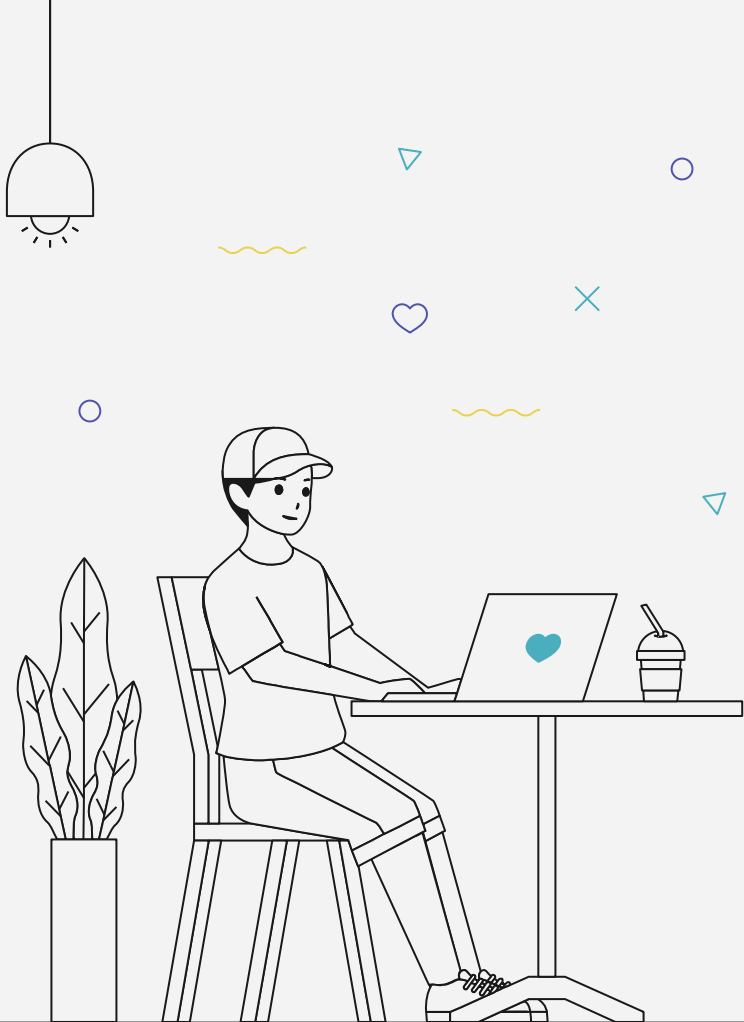
Top 5 drugs: country-level sales rank

| RANK           |  | 1 | 2 | 3  | 4  | 5+ |
|----------------|--|---|---|----|----|----|
| COUNTRY   DRUG |  | A | B | C  | D  | E  |
| Australia      |  | 1 | 2 | 3  | 6  | 7  |
| Brazil         |  | 1 | 3 | 4  | 5  | 6  |
| Canada         |  | 2 | 3 | 6  | 12 | 8  |
| China          |  | 1 | 2 | 8  | 4  | 7  |
| France         |  | 3 | 2 | 4  | 8  | 10 |
| Germany        |  | 3 | 1 | 6  | 5  | 4  |
| India          |  | 4 | 1 | 8  | 10 | 5  |
| Italy          |  | 2 | 4 | 10 | 9  | 8  |
| Mexico         |  | 1 | 5 | 4  | 6  | 3  |
| Russia         |  | 4 | 3 | 7  | 9  | 12 |
| Spain          |  | 2 | 3 | 4  | 5  | 11 |
| Turkey         |  | 7 | 2 | 3  | 4  | 8  |
| United Kingdom |  | 1 | 2 | 3  | 6  | 7  |
| United States  |  | 1 | 2 | 4  | 3  | 5  |

FIGURE 4.15 Use color sparingly

04

# CHART TYPES

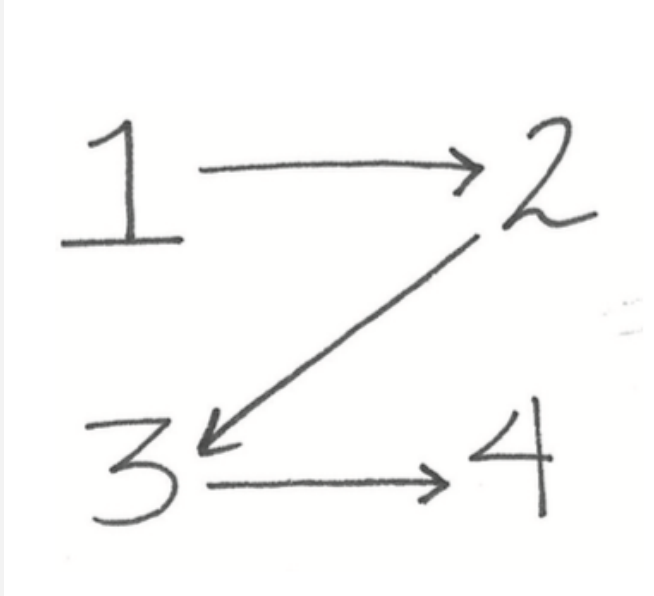


# Choosing an effective visual



\* Nguồn: <https://www.storytellingwithdata.com/chart-guide>

# Position on page



- The zigzag “z” of taking in information on a screen or page

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# Thanks!

**Do you have any questions?**

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